

2023 Video ReviewStream ~ Active Inference Institute

Source: [YouTube](#) Playlist: ReviewStreams Duration: 1:03:56 | Uploaded:
Unknown Transcript method: auto_caption

hello and welcome it is December 22nd 2023 and we're in the 2023 active inference video review stream so welcome to the active inference Institute we're a participatory online Institute that is communicating learning and practicing applied active inference this is recorded in an archiv live stream so please provide us with Fe feedback so we can approve our work all backgrounds and perspectives are welcome and we'll follow video etiquette for live streams head over to active inference dorg to learn more about live stream projects and other projects and groups today's going to be a fun stream so thank you Dean for joining the goals on the slide are reflective learning discuss deriving the 23 set of posterior probabilities and look towards the 24 anticipatory set so maybe by way of introduction could you go a little bit into that Dean well we we always find ourselves at this point in the year thinking well okay well have we closed the loop and and are we opening up another one and that's that's something that uh I kind of look forward to because it's not just a sort of a moment in the calendar it's also a question of whether or not we see what we've done in the in the in the previous lap around the sun we see ourselves in a different way and I think we well I won't speak for you Daniel but I certainly do every time we we reach these uh moments uh I I I kind of look forward to looking back yeah I as well so big questions what was our stream of attention in 200 23 we have courses mechanics and guest dreams Galore in the so very dark Imaginarium so what does that mean for you first um well there was there was a certain presenter at our Symposium this year that her her talk was about a a dark Imaginarium and and you know what would that be like what would that what would that space look like and how would you quantify it and so I think one of the things that just the sheer variety and the ways that information was shared this year um spoke to the fact that there is no one way of seeing things or presenting things or making things available and so that's why I meant by so very dark imaginary okay nice and then where and how will we stream in 2024 more new series voices formats and languages continuing and developing the forms we know and love more podcast and online formats and revisiting materials with the actm journal what else would you add or maybe let's leave that for later let's jump into the main um triptic let's jump in there and then if

you're watching live let's return at the end the most of the stream is going to be going through each of these series just kind of one slide per series and reviewing the 141 videos that were prod this year and uh we'll return at the end to where we want to go in 2024 and whoever's there will be included in 23 um yeah there were 141 videos overall here's the sections that we're going to go through and the products here are placed into the inbox of the active inference Journal where we have utilities that are transferable outside of our Corpus and then also appi to our Corpus it results in the active Journal repo and so that's where we work on the transcription translation so a story for another day but thank you to everybody who works on that side of things so any comments before we jump into the series lots of different ways of creating a paa even though it from a a formatting and structuring uh standpoint we all we all never know I don't think at the beginning of of our next cycle exactly what's going to end up being in the in the big pot of things and that's one of the other reasons why we do this thing at the end of the year just so we can uh kind of see what the difference is between what we thought we were going to be and what we actually turned out like all right let's do it so we had four roundtables these are Institute scale up meetings there are four of them they were informative we hope and Dean you wrote keeping the pie and pizza explain yeah we just the fact that the round table seems like an appropriate metaphor the um the whole idea of making sure that how we communicate with the world outside of the Institute I think really matters and I think you guys do an amazing job of making sure that anything that you believe um will make it easier for people who may not currently be engaged with us to find out who we are and maybe get a better sense of what we're about um you guys work on that really hard and appreciate the fact that um not everybody gets what the active inference stitute is but you guys are certainly pumping out as much um dialogue and and information on that as you can thank you I was thinking of the joke like oh please cut the updates into four pieces I'm not sure I could eat eight all right into the main content so leading off we have the course with Chris Fields physics is information processing and with Andre rir as the assistant this was an amazing series there were six lectures that were inter woven with discussions and interactions Dean what can you say well first of all thank you to to the the three primary um people who spent it again just a an immense amount of time preparing for this and making it available to people free of charge I think it was one of those moments where the active inference Institute could have and Chris and yourself and Andrew could have easily questioned whether or not there wasn't some value that you couldn't have been able to draw from this but I think the fact that you put it into the common space and gave everybody a chance to be able to have a look

at this uh just speaks to the fact that um there's good people who have lots of good information and open themselves up and make themselves available so thanks to three of you and then everybody else that joined in at various points yeah and really it just represented an already consolidation of some form of the quantum free energy principle that Chris fields at all have been working on only in the last several years so that was already cool to see and the second course sounds like a meal already the second course um of the year was the active inference for the social sciences constructing cultural landscapes so this was a great collaborative work with AEL gwin and Caro Ben White Mal abasan and lorenzza ganela myself and AEL kind of had the vision for this course and it was co-organized with cyos research each of the people listed had a topic so we kind of roughly blocked it out that way with ail's vision and collaborating on how we wanted to consolidate and bring together some of the threads of discussion around active inference in Social systems we had AVL with an introduction Ben basics of acadm an incredible lecture I did a section on Collective Behavior Lena on semitics and semantics Ma on Norms and avel with constraints so this was really fun and we hope that the people who joined the discussion enjoyed interacting with it it just felt like the opening of a lot of increased interest in active inference for social systems and just the beginning of the discussion on it but really cool to work with multiple co- teers and do this second course using some of the templates and lessons that we were learning from the chrisfield side all right we had the third PL active symposium enacting ecosystems of shared intelligence and this was really fun there were a whole team of co-organizers so thanks to everybody who did that and all the guests were listed on the side there were two intervals the program still up we had a great Round Table what do you think about this Symposium or what would you add I think you must have had to drink a lot of coffee because you you're kind of in the thick of this uh of this weather system for a long time I think it was amazing I mean the the the support from our presenters the support again from the people who made sure that everything sort of came off with a with a reasonable amount of consistency and Co coherency um again if you want to if you want to be able to get a real clear sense of how much change there's been going on in the last few years and how Dynamic the understandings around active inference are simply open up one of these symposiums and just get a get a sense of how how much the spectrum of research is continuing and evolving yeah lot to say on that too but it was a fun topic and a really memorable Symposium with an application focus and also I remember Pablo presenting the first AC game and us being in the the digi cyber physical cognitive ACM game space and a lot of other memorable moments there there was the dark Imaginarium there which is still one

of my one of the greatest fantastic moments to listen to that that was awesome yes all right well Darius did incredible work in the last several weeks with bringing out the first five episodes of the insights Series so we had Carl Friston, Marina, John, Verity, Hippolyto and Marilyn Stera just in the last several days honestly each of these were incredible I learned so much from returning guests and from new guests and it's really fun Darius thank you for stepping up and for doing a lot of the prep and having these conversations that that speak for themselves so it's really fun to see any other comments on that Darius says I don't know how what his background is but he's got a face and a voice for podcasts for sure nice all right into the series we go morph Stream So Sarah Hamburg over here proposed as part of the scientific Advisory Board and just all around active friend proposed this neuromorphic plus active inference combination stream and so far there have been two morph streams shown here and it's really been cool and it will be fun to see where this goes as the neuromorphic and unconventional Computing spaces develop and as active inference is used in the measurements and modeling and design of unconventional Computing and synthetic intelligence this is going to be a really exciting stream so thank you Sarah for leading these discussions and reaching out to people and everything that's something weird there morphic what is it that a neck beard that you're sporting in that picture like what's did you morph into something what happened there not sure is this been is this modified is this is this I'm not sure oh wait a second I was like wait a second you took this really seriously I barely know what to say next hilarious well played sir um we had orc stream with Shree Jane on Communications and secrets in organizations and with Anamarie Swan on the ecological organizations framework and a really interesting and relevant topic active inference for and with and by and in organizations and in ecosystems so it's really fun to have these conversations people who are in different places coming to active inference and having them share what they're working on and how that relates to organizations and I know this will also be a fun Direction next year anything that all right math stream we had three of them we had Christian Bodner on topological deep learning graphs complex and sheaves Shan Tol on active inference and string diagrams with Ali and Vincent Wang, Masan on Constructor Theory as process Theory with the very memorable chalky Dusty board and absorbent jacket definitely a Time series and a work of art and motion but overall these three not to say that math didn't come up in other streams it did but just kind of looking at these three it's kind of cool to see the discrete and the topological and the categorical the applied category theory side with Shan Tol that was also there in live stream 54 which we'll come back to but those are some developments that are already being

applied to active inference and some that are just kind of one step away so a few of those discussions started to forge some paths that are really exciting for the math of active inference yeah when it gets to the sophistication that these these folks are able to bring to it I just had to tap out there were I mean I can get up to category Theory and and at least fake my way through some of it but I was I was amazed at the uh just the confidence that some people that are are bringing to this this whole idea of where where it can it go yeah it's incredible all right model stream so these are streams that focus on the generative models and do some kind of Step through with models so we had model stream 7 the second part in the pmdp series here with Carl Adam Connor and yakob fun discussion and really applied there model stream 8 with Tom ringstrom on reward is not necessary part of a really fun Technical and broader discussion on what's the role of reward what's the role of goals surprise all these different cognitive phenomena which ones entail one another which ones are super sets of one another and this kind of model uh without reward that Tom proposed and then model stream n with aswin Paul on efficient computation and active inference also a symposium guest but aswin and at all have been doing some really awesome work with a computational efficiency and kind of tractability of applying these models so that's really cool to see all right textbook groups so there were three cohorts that had some activity this year there was cohort 2 which finished the second part of the book second chapters 6 through 10 in the beginning of this year cohort 3 which was entirely within this year both halves and cohort 4 uh or sorry someone finished it half someone started half one of them was complete cohort 4 was the one I guess that was completed in 2023 but the dates really don't matter we did a bunch of textbook groups with this 2022 par at all textbook and it was really inspiring there are so many ambitious and excited Learners who come 0 1 3 25 times and people engage in so many different ways with the synchronous and the asynchronous and people who just want to listen have the recordings so that has started to develop really something special in being there and just appreciating everybody what they bring to this book and also Ali in particular for really bringing a complimentary set of very exacting accurate understandings of what is in the textbook and also very broad understandings of what's not in textbook we had the book streams so Ali and I did some book streams on the textbook also with Tyler and blue we did a series on Bon hz's book governing continuous transformation reframing the strategy governance conversation so this was pretty fun to go chapter by chapter with Tyler and blue and talk about what does free energy principle and active inference mean for organizations and talk about just a lot of cool topics and have B John join for some of the discussions too yeah any other comments on the

book streams just to just to thank you to the uh authors when they were able to uh to join us because I think that again it brought the book To Life yeah thas Ali because when whenever Ally comes and and joins us you can be sure that he's going to he's going to bring his aame he never comes without being the most prepared guy in the room so that's saying something because Daniel likes to be well prepared so very true very true but these were fun okay now we'll go through a couple of slides with the guest stream so this is a really broad category um it started out in January with Adam peace talking about Sumo and neuros symbol IC text Ali Resa modish talking about the 18 subtypes of surprise avel sharing some work on the physics of creation then a very fun and memorable conversation with Matt Bowski and Jordan Hall J Benjamin falland days talked about gibsonian ecological psychology and resonance Baba Brinkman shared a lot about the artistic process and did some freestyle rap played music on the stream which was really fun Max Berg talked about rumination shui at all talked about richness and ineffability and this was a philosophy with some technical details uh van Visa talked about language models and Consciousness Elliot Murphy and Steven pend DOI had uh at times contentious discussion on language models and Chomsky that was quite an event and ju Diego Bata and Zacharia jebber had a conversation shown here with Dean in hat and Ali on time Consciousness and computational phenomenology so any of those that you want to comment on well well there's a few I thought that I thought that Jordan and Matthews um you call it a dialogue they I think they were trying to understand one one another it was uh there was a bit of a c circling going on there but I think they they kind of left with a better understanding of each each person's position Baba's was I never anticipated what where that was going and I just I loved it by the end I was just like wow there's there's some people who who absolutely can uh can see things in ways that really open up your your idea of what a possibility is and for me one of the ones that I really enjoyed was um participating in the on the time Consciousness um question I found that any time that people will place an emphasis on change and dynamic change to go along with some of the more stable measures that that kicks out um I want to sort of figure out what they're pointing at and so I thought they did a really good job of um trying to explain where where they were coming with that I actually made connections between the stuff that they were talking about and the um stuff that Shanah doson talked about in the Symposium with the dark Imaginarium again you have to kind of see them Side by each but where they're talking about how we divide time into objective time and subjective time and how you're in the moment and how you can see yourself as being having to be outside the moment exactly at the same moment and is that a contradiction or is that

complimentary I just thought that um there was a lot of that possibility this year but again you don't necessarily see people coming using the same language or even using the same Quantum frames and yet they're still talking about the same thing if you can put the prepare and the measure up on an equal footing and let those two things work in concert uh it seems to draw out of people this oh my goodness have we thought about it in this way so that was what I saw from this first segment of the three parts of the guest streams this year awesome all right guest stream part two so Tom froze on eruption Theory really cool part of a bigger discussion about Free Will agency all those topics Bobby aarian coming back for 15.3 on the theological stance and building that road to Omega uh suchia and saigo on category Theory and Consciousness and integrated information Theory this was really fun it's always exciting to see category Theory drawn out in real time and what people show as they construct and Trace so that was really fun and I think it also triangulated with a lot of discuss that we had been having about information information Theory integrated information Theory all this discussion about Consciousness different kinds of models um Maxwell and Ma joining to share one of the Crossovers with Quantum information Fe and Consciousness on the inner screen model with with some other guests and a super fun convers ation Denise Hol active inference AI in the spatial web symbolizing a year of excitement around the spatial web and various other developments in the active inference EOS system and how that was in June 2023 how that is in December 2023 Sam bini leche on predicting reflame River reflecting framework for dual process Theory more large language models and cognitive science and action Adam saffron and Arthur gilani on deep canals was a fun discussion with a lot of mapping and territories in the brain Megan Peters and Nora Bradford having a very interesting conversation about Consciousness research and how it's communicated and carried out so very complimentary to a lot of the other discussions ongoing beatini and C Yuna discussed selfless mind unlimited bodies bringing together some of the basian phenomenology with meditation and different ways of knowing salvatory causal inference via predictive coding the uh elade and Dell G gestring on the backprop free 4D ant-based search pretty fun representing kind of like The Cutting Edge in ant algorithms which is always a fun opportunity to talk about ants and active and Alexi to chinsky who led this discussion of one of his recent papers a case for Chaos Theory inclusion and neuro psychoanalytic modeling and just brought together this panel and so that was very memorable because there are different voices just it was like the paper being released and just having a participant in our learning groups and part of our live stream series and just fostering the discussion just seeing how rich and how how people see

these works that are coming out month by month so any comments on any of these streams no in the middle of the year I was being a carpenter and I was also trying to make sure that I could say something coherent about category Theory and Chris Fields stuff so I wasn't I wasn't present for most of these fair enough all right then the last third of the guest streams so Andres Gomez emilson and Chris Percy on the electromagnetic field topology Consciousness so this is the kind of qri Consciousness qualia research very fun and also really provocative Pang and fornito on geometric constraints in human brain function slow traveling waves in the brain uh Gregory Sergent on agency with structured latent spaces on the math and phenomenology side Andy Keller on neural structure and artificial neural networks Carl friston Chung and Schwarz joined for a very interesting discussion on working with Gerald Edelman who won the Noel prize and was a very uh influential researcher in the neurosciences and hearing from each of these three fellows their experiences working with Edelman was incredible and it was really historic and gave a lot of context on the theory and the people and I hope we can do more they were impressionable young men that's what they revealed they shared about that time in their lives and that time in in how it was and and uh I I I look forward to more of this working with format so that was pretty fun um Kristen Whittle Reflections and implications of a naive attitude to architecture very beautiful presentation and and far-reaching and and Majestic IC and touching upon a lot of aesthetic comments and um very poetically addressing questions about applying active inference in practice not applying it in theory in practice um David Tucket conviction narrative Theory narrative and active ever green topic Ramen on myth of objectivity and the origin of symbols going a little bit deeper from The Narrative into the symbolic Michael Carl uh great textbook group participant presenting his research on deep temporal models of the translation process which led to some very interesting discussions about the ey moving and the heat the keyboard typing in the translation process Mel pakari gave a very philosophical presentation on incorporating variational free energy models into mechanisms the case of predictive processing under the free energy principle and I'm I'm a big fan of this presentation and paper and here's why when people are giving mechanistic accounts for different physical material systems the the idea that um one would doubt the relevance of a material physical flow is almost like heresy like of course the windmill works because of the flow of air or however one wants to say it but the flow of gas gasoline the engine things like that and yet here we are talking about free energy flows so framed in many ways we have this question that sometimes we take a stance on or not is what is the status of free energy flows is it just an accounting device that's just epip phenomenal or does it have a kind of causal Force

within an explanatory account just like the flow of water downhill power this energy generation could it be said for some or which systems that flows a variational free energy play an actual causal explanatory account and in what situations can we only pull back to say well this is playing a role in my surveying of this situation but not necessarily playing a causal role in that territory but yes it plays a computational role on my map that's the kind of instrumentalist stance so this was very deep work that validated and connected those discussions that are ongoing in the active inference back to broader threads in history of science um Elliot Murphy talked about Rose a neurocomputational architecture for syntax maybe we'll come back to that one Adrien Bon with Susan Hasty on the physics of Life how to predict Evolution a fun and far- ranging conversation Phoebe Cseto and Dan Simpson on towards basing World models was really cool talking about some uh current trends in thermodynamic AI and causal World models and then the most recent Gest stream with Andreas Kording on the logic of model evaluation of noisy judges which was also very memorable in probably multiple ways so what which one of these guest streams do you have any comments on well one of them one I want to just shout out to Andreas for being so passionate and being so confirmative he I just love the way that when you and I were interacting with him he was just like yeah you get it you get it and it was just it was really cool to see somebody who obviously has capacity that are amazing but actually when he can have a conversation and an interaction with people who get what he's obviously you know invested a lot of his time and his passions into when he can have that conversation with other people who understand and can converse with him on that that was just that was I thought that was amazing so yeah it was very fun all right we're gonna do one slide for each of the five paper live streams and then that will be the end so we'll do five more slides one for each of these live streams then look at the live chat see any questions or topics and kind of talk about Reflections where we go okay so earlier in the year live stream 52 geometric methods for sampling optimization inference and adaptive agent by and dasta at all and I prepared the zero and had a one and a two with Lance this is some of the clearest most updated most concise descriptions of some of the signal processing and control theoretic and basing mechanic aspects of active inference and the free energy principle these were super informative the kind of motif or image that stays with me is this figure with the ball rolling downhill which we talk about all the time path the least action free energy landscape most likely thing happening Bayes optimal ball rolling down the hill and then it's falling downhill here and it could be like through air or through water or through honey and so

different acceleration Dynamics on that bowl and then different ways that you can use the statistics of the acceleration to jump ahead and do an accelerated optimization by jumping ahead not too much or not too little so kind of using the computational resources adaptively so that you sample and this was just a really great discussion so big thanks to Lance for joining timing acceleration in Geometry add water and you'll have a fantastic Brew try at home yeah okay 53 one of probably the most memorable moments of our life Dean to be real yeah yeah um with uh the set of papers snakes and ladders in paleoanthropology and to copy or not to copy with Walker manrique and friston and Dean and I had a multimon Foothill Journey from the base camp to the do one and Pierce the Veil with a DOT zero in the side by side by side by side and we had the dot one and the dot two with Michael and Hector Carl joining so what can you say about this series well first of all our guests were amazing Carl Michael and Hector Inc again people who are really really invested in their craft and really really are not ready to sort of just be herded into a certain way of thinking being uh adopters now of what I think Michael and Hector see as something that can really open up spaces which is act an active inference lens on their work in compared to psychology and anthropology and archaeology I want to shout out right now that I think one of the one of the greatest gifts that they provided us was allowing us to look at two of their papers concurrently and I think that next year one of the things that I really am going to try and um convince people of is if when not just if but when we do live streams if people that have authored papers would be prepared to come in and show two two or more of their papers at once because it really helps us show the potential of looking through an active inference lens at work so yeah it was very memorable it was it was lovely we've we've actually formed social bonds as a result of the commitment that they uh have shown to the active inance community and they didn't come you know with a with a background in the active inurance Community but when we reached out to them they reached back with open arms it was really amazing yeah all right live stream 54 mathematical foundations for a compositional account of the basian brain by Toby St Clair SMI yeah well first a mega shout out to the whole team many synchronous and asynchronous foragers who met weekly for all of 2023 with many incredible persistence and contributions nonethe least of which Jr who absolutely blazed the way forward with an open source contribution to a category Theory AC of inference Adventure story curriculum narrative journey and um the week by week as we went through this um PhD dissertation the week by week and the the journey and the 100 foot waves that we cited and sometimes got swept away in just in the months leading up to the conversation with Toby and then to have the zero here with Ali but

again building on the contributions of many people and the months leading up to it to have the do zero and do the impossible to bring the dissertation and the six months into what can you really say about o infant category Theory which is like he said what there is to say in the dissertation so what can we say about what there was to say so that was such a fun question and then to have Toby join and have just such authenticity and um relatability with how he approached category Theory so yeah what would you ask I was gonna say I think that I think that the way because I I mean a person of Toby's intellect that could be able to write at the level that he wrote for his dissertation to also be so approachable is is I guess I guess normal for him but I would not I would not have anticipated the approachability that Toby had the other thing is is that I I cannot thank Ally enough because the number of times that we were having our conversations as we prepared for this under wave uh the number of times where I just felt like I I can't be here I can't be doing this because I got I got into the first three sections of the paper and it was and it was quite quite literally so it was so unbelievable the amount of stuff that was packed into this thing I thought I don't know who the reviewers of this are but good on them yeah I I I Echo that thank you Ali actually showing up to connect the dots and just kind of be like okay so here's the thing with the funter we needed that otherwise we were off the rails I mean honestly I can I can appreciate what NASA does now when it sends something off and it knows that it's going to meet the asteroid at some place hundreds of thousands of miles away and blow it up because that's what this felt like this felt like I was trying to find some dot in some distant land that I had to land on and it was Ally and and you and and then of course Toby when he actually showed up and it was so chill oh amazing all right live stream 55 realizing synthetic active agents the triple play in the title and this was a great Series so first big shout out to yakob and Bert and also some others but for helping with a do zero and for us really coordinating to prepare this material because this is kind of another impossible compression which was to bring together the message passing on the on the be constrained free energy on the all of these details that just seem to stack and stack and stack and bring us closer and closer to The Cutting Edge but then Magnus being so chill so clear in the first discussion in the second discussion showing the code so sparse so tur and just being very friendly and willing to continue to like help us and learn how to apply generative models with the RX andur toolkits and Julia and to make what literally looks like something like a circuit diagram or a glass bead game or some other kind of esoterica what makes this have at least a touch point with our understanding of figure 4.3 or figure figure 7.3 from the textbook and just the ways that the basil motifs of articulation of sense making and action and policy

and attention all these motifs that we kind of explore and the base graphs and the forny graphs that that entails all of this and then to connect it to the computational advances like the direct policy inference so instead of unrolling policies and then like sampling across them to do this kind of direct clamped unclamped direct policy inference on the generalized free energy it's happening so fast that it's incredible and it was just really great that Magnus at all brought it out so comprehensively and that magnus's generosity in also wanting to see this be really adopted because it's like a toolkit and a framework that is being developed in a capacity that's very fundamental for our ecosystem not just some Claim about some system of Interest a very casual Observer I think Magnus could translate anything into a map or a graph anything like he could translate meditation into a graph or any and again that's from the Casual Observer if it again I in my very colloquial sense say gripper and gripped and when I was watching you guys go through this that's exactly what I was seeing and he was I mean he wasn't using my language but he was certainly showing how it can be done all right and then the last live stream was 56 so here kind of embodying Dean's exhortation that authors request a minimum of two papers Alexander aoria dispatched four papers to Blue and I so we scrambled through to each so thank you blue for being very generous with with with working together this this was fun um and then Alex joined for the one and the two we talked about the common model for cognitive systems and about modeling diverse intelligences and then in the second discussion the 56.2 he took it in the Mortal computation direction that was like a paper that had out on pre-print in the time between the one and the two so it was just incredible that was like in the one it was a paper that hasn't come out yet but oh it'll be cool which just from like a live stream perspective I'm always like okay that's cool but by the time that most people listen to this it will have been released so it's always cool to have somebody share that but at the same time it's kind of like it makes the work more externally reference IAL instead of containing the material which is not a bad thing it's just how that kind of speech Act is and then for then the second discussion the prepare measure with the real world getting in between the openings and the cracks and the door is just awesome so this was fun and uh Alex has a lot to share and it's just exciting to hear about all these like relatively very technical detailed cognitive model architectural motifs play into bigger topics and questions okay all right well here we are reflections so we have a few different things here we'll probably write a few different things down but this is the last slide so people watching live write any questions or comments and then we'll look through it in just one minute but first in where does this take us or how does this reflect upon us now well again if I were to if I were to say what would 2023

if I was to pick one thing and say well that that explains 2023 I think fact that we are talking about Reflections we're looking back over time and I think with what Chris was able to share with us what our many of our guests were able to share with us the fundamental nature of how we describe time how we use time how we prepare and measure for time what Quantum reference frames mean in terms of change Dynamics um I think all of those things now seem to be a part of the Lexicon whereas before we would kind of we would kind of go out look at it a little bit and then come back to whatever we felt we were more familiar with um I don't know what that means for 2024 other than I think that if we if we anticipate that we really still haven't um move to a place where the the nature of how we apply active inference perspectives on the things that we do we're still in the very nent stages of that I think that I that the um the way that we incorporate our understanding of the change and the Dynamics and the timing not just taking our time but because I believe that active inference really affords strategy how we time our takes is still in front of us it's still something that I think we're going to see a a lot of development in and I think it's going to be really exciting for people because I think up until now people are still just trying to get a grip on what is active inference they're doing a lot of the measuring things but I think that when we actually see the change being and the and the and the ability to to understand the change the carrying capacity for that I think there's going to be a lot of people that are going to jump on that perspective and and run with it and are going to want to really share what that what that means in terms of the the things that they do and the hopes that they want to try and um realize so that's my sense of what this year was and it was I think it was quite a bit different than yeah yeah 20 22 just from the top of my head it felt like there was more developments in basian mechanics narrow that was kind of the basian mechanic um miraculous year of developments with more paper Focus discussions yeah heav Kronos in 22 yeah then we we kept the chronometer up and also brought in this guest stream the diversity and people who had um more and less familiarity with active inference and higher order discussions about the role of technicalities and a formalisms in philosophies and um the continued simmering and marinating of the um what does this mean question multiple kinds of what does it mean what does this mean for people what does this mean for humans what does it mean for sentients and cognitive systems what does it mean for the planet what does it mean for all these different and then [Music] um category Theory painted some incredibly promising lines on the physics that was brought out in 22 we had like the streams and the videos presented here this isn't a summary of the ecosystem so a lot of interesting things happens and we we only sample arbitrarily truly but one thing I really look

forward to in terms of like looking back to look forward I really look forward to working with Darius and others to integrate the podcast and live stream into the production and have an integrated process for reaching out to people and scheduling the right kind of event and the right kind of activity and then just having a very professional very Broad and deep and multilingual channels and have fun and plug it into the journal and just like have the conversations have the right signaling and the wisdom to know what is going to be public and language models and transcriptions and translations and what are the conversations that aren't that way and what's like the bigger picture also one one kind of image when we were just before we started and we talked talked about like how different the 0.1.2 One 2 3 tap tap tap is and about how the multiple touches is very different than the kind of one and through or like one and out of a guest stream and that kind of one measurement that appears on the screen versus that repeated contact and the a kneeling that happens there so yeah what do you think about that well again I I I I you know it's like your kids you're not supposed to pick your favorites but I do anyway I but the way that way that Chris spent the time with us on the discussions as much as he presented the way that the the course of the interactions in the guest streams where people had their presentation but they also made sure that they had time to actually have that conversation I think I think that's very powerful and very valuable I know it's time consuming but again if we're talking about whether or not we want to be efficient or whether we want to be effective um we're just I think we're really glad that people are prepared to stick around and make sure that that not just for for clarification purposes I think what I'm finding is is that people that are now engaging with us are find in that they're able to both share and be shared with like it doesn't feel like it's a unidirectionality if we have nothing done nothing else I think we've proven out that we want to be a community that we want to create an ethos and that we want people that want to participate not just present but actually get in get in the get in the ring with us and wrestle with this stuff because I think as Andreas pointed out in that last guest stream um we take on really hard problems problems that maybe can't be answered in in a 15minute presentation and I think that that serves a real purpose so again if it's if it's about as we grow as an as an Institute what we and who what and who we engage with and for for what purposes I think at the end of the today we continue to try and find as many different ways of organizing that so that people can come in and like I said move past the instructional you know what why that's efficient and move on to the the uh interactionist because I think that's where a lot of the stuff you want to interact with our library by all means come in and do that and then come and talk to us and let's let's find out what

value that's that's serving and how we might be able to distribute that into a a larger a larger space or a larger Place nice all right I'll read some comments from live chat so upcycle Club thank you Joanne thank you thank you Nathaniel all right thank you Susan I'm going to read your comment I like what Dean said me too I'm paraphrasing putting the generative model to work can we expand on that and what realistic metrics objectives might in that regard be for the institute for 2024 aspirational outcomes for 2024 what will be the challenges both for ACM Institute and possibilities for using actim to advance the Evolution for AI in general I wish how do we how do we think about challenges I mean do challenges even exist and actin is something that helps us address them are challenges something that are within an active gripper and gripped relationship with the work does the otim perspective reframe what is called a challenge by other words well I think one of the things that we've talked about in you know in a variety of different spaces is that there we're going to we're finding ourselves with with things like Grant applications being asked to meet certain standards and I think we're quite capable of doing that but I think one of the things that we're talking aspirationally is not not just setting ourselves up to meeting a minimum meeting a standard but actually trying to cross thresholds and Boundary cross and move into spaces that simply are not being occupied at this point now that's not we're not colonizing anything but we are acting as pilgrims and maybe there are thresholds that we want to get over and not just barely get over but maybe really truly kind of break away from some of the gravitational pull as long as we can get back right safely but but that's one of the things that we're talking about right now not just necessarily meeting standards as that form of challenge Ares but also surpassing maybe by orders of magnitude so we'll see what's this image mean to here oh I just like things that are orthogonal um and looping because I'm a big fan of learning loops and not necessarily unidirectional so um whenever I get a chance to reflect I can look at things um bidirectionally and so this now I get the luxury of being able to look back looking back isn't linear it's very non-continuous I can hop around all over the place I can think of 55.1 and I can also think of uh the Symposium and certain presenters there that had the first game the first animated Game and thinking wow now of course if I think thought about it in a linear sense and everything was sort of sequential that would give me one image but now that I'm thinking back on it I get a whole different idea of of just exactly what that as I say that lap around the Sun generated for me yeah in that Spirit here's me in the passenger seat looking forward but taking a picture that represents back backwards but backwards coming to the present all in the forward-looking moment all captured on the screen all right if anyone has any other comments I'll just wait like a few more

seconds otherwise we have a few I want to thank the officers I want to thank the board I want to thank the Sab this you don't have to you don't have to play me off that's that's it I don't need to get into some crazy political statement but again I think this is a a perfect time to express my gratitude for all the people that do all the hard work thank you I Echo that just speaking to the streams and the videos I really hope if people are interested in learning these skills of audiovisual production getting involved with using some of the automations and tools that are out there now and helping like Leverage us to have better and new kinds of materials hash con content creator and Catalyst and just kind of Enthusiast and um co-learning all right thank you Dean farewell thanks everybody bye